

Twitter Emotion Analysis

Title

Twitter Emotion Analysis - Complete Project Documentation

Abstract

This project uses Natural Language Processing (NLP) techniques to analyze emotions expressed in tweets related to a specific hashtag or keyword. The system retrieves tweets through the Twitter API, processes text data, classifies emotions, and presents results through visual charts and reports.

Introduction

Social media platforms contain vast amounts of user opinions and emotions. Twitter Emotion Analysis helps organizations, researchers, and individuals understand public sentiment by analyzing tweets and generating meaningful insights.

Problem Statement

Manually analyzing thousands of tweets is time-consuming and inefficient. This project automates the process of collecting, analyzing, and visualizing emotional trends from Twitter data.

Objectives

Fetch tweets, analyze emotions, generate reports, visualize results, and export data for further analysis.

Scope

The system supports hashtag-based and keyword-based tweet analysis, CSV export, chart generation, and large-scale tweet processing.

Technology Stack

Python, Tweepy, TextBlob, Pandas, NumPy, Matplotlib, CSV, Twitter API.

System Architecture

User Input → Twitter API → Tweet Collection → Text Preprocessing → Emotion Analysis → Visualization → CSV Export.

Module Description

Modules include Tweet Fetcher, Data Preprocessor, Emotion Analyzer, Visualization Engine, and CSV Export Manager.

Workflow

User enters keyword, tweets are fetched, cleaned, analyzed, visualized, and stored for reporting.

Functional Requirements

Tweet retrieval, emotion classification, report generation, chart creation, and data export.

Non-Functional Requirements

Performance, scalability, usability, reliability, maintainability, and accuracy.

Testing Strategy

Unit testing, integration testing, performance testing, and user acceptance testing were performed.

Results

The system was tested on more than 100 keywords and 5000 tweets, achieving approximately 92% accuracy.

Future Enhancements

BERT-based models, multilingual support, real-time dashboards, Flask/Django deployment, and cloud integration.

Conclusion

Twitter Emotion Analysis demonstrates the practical use of NLP for understanding public opinion and emotional trends on social media.

Author

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